

Assignment-8

Q1- What is Data Cleaning?

Data cleaning is process where we clean data for train models means ML needs structured data(clean data) to train models. So in simple words data cleaning is a part of data science where in remove unstructured data like – missing values , duplicate values and outliers(abnormal data). From a millions of data after cleaning the data we use it in ML to train models.

>Missing values- Loc , Price , Square foot.

PTK	15000	10
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Surna	15000	12
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Bungal	NAN	15
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Badhani	25000	NAN	{In Bungal and Badhani data is missing}
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>duplicate value- name marks

1. Krish	55
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2.Parav	65
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3. Krish	55
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4.Aryan	55	{ 3 rd one is duplicate}
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> outliers(abnormal data) - Data= 10,11,12,14,16,17,19,20000.

{ 20000 is a outlier}

Q2. What are missing values?

Missing values are incomplete data in total data which we need to clean or modify. Like we have millions of data in which few ones have incomplete data means missing values.

Eg= - Loc , Price , Square foot.

PTK 15000 10

Surna 15000 12

Bungal NAN 15

Badhani 25000 NAN {In Bungal and Badhani data is missing}

Q3. What is NaN in Pandas?

NaN defines not any number basically it is a missing values in data.

Eg- Loc , Price , Square foot.

Bungal NAN 15

Q4. How do you check missing values?

We can check through this the code which given in below.

```
import pandas as pd
import numpy as np
data = {
    "Name": ["Aman", "payal", "krish", "parav"],
    "Age": [21, np.nan, 20, 23],
    "Marks": [95, 90, np.nan, 85]
}
df = pd.DataFrame(data)
print(df)
```

output-

Name Age Marks

0 Aman 21.0 95.0

1 payal NaN 90.0

2 krish 20.0 NaN

3 parav 23.0 85.0 { NaN is missing values}

Q5. Difference between dropna() and fillna()?

In simple words, Dropna() removes incomplete data, while Fillna() completes the missing data by replacing it with suitable values.

Dropna()- It is used to remove rows or columns that contain missing values (NaN). It is useful when the missing data is not important or when removing it will not affect the analysis much.

Fillna()- It is used to replace missing values with a specific value, such as 0, mean, median, or any custom value. So data can use to train.

Q6. What is duplicate data?

Duplicate data means in total data there is some data repeated,

Like same value and name repeated after some data in total data.

Eg-

	name	marks
1.	Krish	55
2.	Parav	65
3.	Krish	55
4.	Aryan	55 { 3 rd one is duplicate}

Q7. What is duplicate data?

We find duplicate data using command `print(df.duplicated())`

Which output in terms of true or false where true means data is

Duplicate and false means not duplicate.

Eg- code ->

```
import pandas as pd

data = { "Name": ["Aryan", "Krish", "Aryan"], "Marks": [95, 87, 95] }
df = pd.DataFrame(data)
print(df)
print(df.duplicated())
```

Output-

	Name	Marks
0	Aryan	95
1	Krish	87
2	Aryan	95

0	False
1	False
2	True { true means data is duplicate in that row}

Q8.How do you remove duplicate rows?

To remove duplicate rows, we use the **`drop_duplicates()`** function in Pandas. It removes repeated rows and keeps only one copy of each unique row. In simple words `drop_duplicates()` is used to delete repeated rows and keeps only one copy of each.

Eg-

```
df = df.drop_duplicates()
print(df)
```

output-

		dtype: bool
	Name	Marks
0	Aryan	95
1	Krish	87

Q9. What is an outlier?

Outlier is also called abnormal data which means data is different from all other data like all other data is 2 digit and from all that one specific data is 5digit data.

Eg-

data- [11,14,18,22,26,30,44,44000]

{in all above data the last one(44000) is a outlier.

Q10. Why is data cleaning important before training an AI/ML model?

Data cleaning is important because AI/ML models learn from the data provided to them. If the data is uncleaned and have duplicate values, missing values, and errors so model may predict inaccurate produce unreliable data. Cleaning the data improves its quality and make more accuracy in results.